Single Cell Bangla Braille Book Reader for Visually Impaired Peoples

Noushad Sojib
Dept. of CSE
North East University Bangladesh
noushad.sust@gmail.com

M. Zafar Iqbal
Dept. of CSE
Shahjalal University Of Science & Technology
mzi@sust.edu

***Abstract*—Braille is a tactile system to represent text. A text in braille consist of several braille cell where each cell represent an alphabet or symbol. A single cell consist of 6 individual dots. Combination of raising high and raising low of these pins compose an alphabet. There is a standard braille code for each alphabet of most of the languages. Generally texts are embossed on paper and visually impaired persons move finger over the dots to read text. Our hypothesis is to keep the users fingers rest over the cell and actuate cell pins in such way that it can make feel like moving finger over cells. In this paper, we will show the design and development of a low cost portable braille book system with only one braille cell along with its experience with the real users. Besides all the available braille devices the challenges was to build such a system in so small space. We tested the device for several months with 10 users. They were able to read through this device and got different feedback from them.**

***Index Terms*—Bangla Braille, Braille Hardware, Accessible Ebook, Bangla Braille Book Reader.**

I. Introduction

Braille system enables visually impaired people to see text by touching over the braille embossed text."Fig. 1", shows reading through braille cell. Each cell represent a single symbol. A single cell have 6 pins and there are 64 possible combinations.There are standard six bit code against each alphabet in most of the language. A braille cell occupy only 2.3X7.5mm area but has 6 individual pins to actuate.There are some commercially available solution to display text in braille format but they are very expensive and do not have suitable way to convert and show Unicode Bangla book as we intended to give an alternative to the visually impaired people to read book instead of costly printed paper system. A refreshable braille display consist of generally 10 to 40 cell. As the reader touch one cell at a time and go for next, we thought to give this feeling using one cell only instead of multiple cells. Like the finger move over the cells, here the reader have to rest his/her finger over the cell and the cell will slide its side in a reasonable time that it will make the reader feel like he/she is moving finger over the cells. The main challenge here was to develop a single cell by maintaining this tiny size and put 6 actuators inside it. We came up with an idea to build 6 actuator within this small perimeter using electromagnetic relay. The following documents presents the hardware design, electronics control system, the software and the experiment with the real users. We got satisfactory response from the blind users with some pros and cons. How good they were able to read and how was their experience reading with this device also shown in the result section.

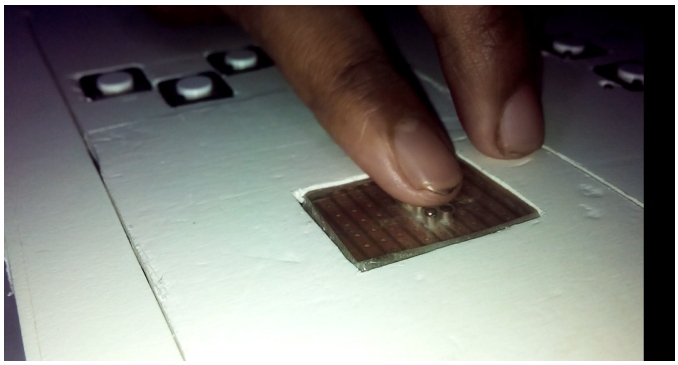

Fig. 1: Braille Reading

II. Related work

There are several works around braille device. As its a globally expensive device there are many local approach developing low cost system across the globe. [1] In a paper [2] author shows some experiments with single cell device. Some authors use piezoelectric device to actuate a single dot [3]. There are several approaches also exist like solenoid, shape memory alloy ... etc. to make linear actuator for a single dot. [4]

III. Bengali text to Braille Code

There are specific code to represent Bangladeshi Bangla text. Most of the alphabet has direct mapping with six pin coding besides some special code sits before numbers and some special alphabets.There is a standard rule to convert Bangla text to braille code [5] [6]. "Fig. 2", shows an example of converting Bangla unicode text to braille representation.

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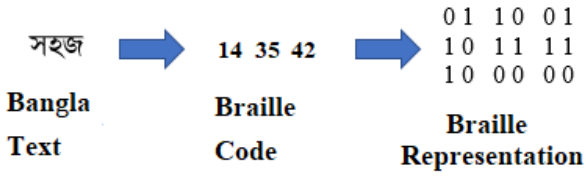

Fig. 2: Bangla text to braille

IV. Development of the Device

Development of the standard size braille cell was the main technical challenge of this book reader. We have developed a software in Java for Computer that can convert any Unicode Bengali text to our specified braille book format. These braille books are then transferred to the Braille book. The device have to read the braille code and move cell pins accordingly. Having so small space to actuate 6 individual pins we have to come up with a innovative way, equally important, low cost of the device. The following sub sections describe its hardware and electronics design. The book reader consists of several part such as the book file format, braille cell, button and sound interface, controller board and the casing.

A. Book Representation

Like a book, with its several chapters and pages, we have developed a file format to represent each book. As the system is developed on an embedded hardware and it has constrain in file/ folder name length, instead of placing whole text in a single file we used a new file format to represent each book. Each book is represented as a directory where the directory name is just a text and the actual book name lies in a text file named "head.txt" inside the directory. Inside the book directory there are directories for each chapters parallel to the text file that contains the book name. Chapter name lies in a text file lies inside the chapter directory name "head.txt". Inside this directory there are text files for each page. Each text file contains contents of a single page converted in braille code. The "Fig. 3", shows the file format of a single book.

Book1
- *head.txt (Contains Book Name)*
- **chapter_1**
  - *head.txt(Contains Chapter Name)*
  - *pg1.txt*
  - *pg2.txt*
- **chapter_2**
  - *head.txt(Contains Chapter Name)*
  - *pg1.txt*
  - *pg2.txt*
- **chapter_3**
  - *head.txt(Contains Chapter Name)*
  - *pg1.txt*
  - *pg2.txt*

Book2
- *head.txt (Contains Book Name)*

Fig. 3: Book File Format

Hundreds of such books can be placed in the storage area.

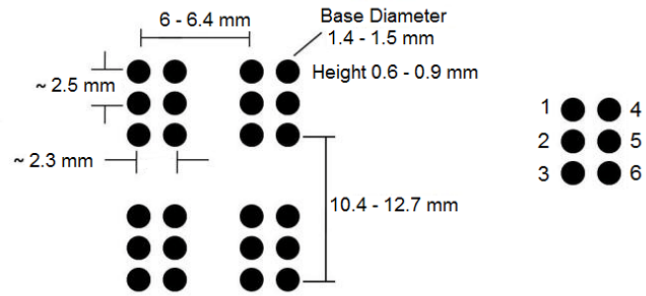

Fig. 4: Braille cell dimension and dots numbering

B. Single cell module

This is the actual hardware consist of six dots that can represent any symbol. Each dots is a linear actuator driven by an electromagnetic relay. The six actuator can independently actuate six pins upward to downward direction. There are six electromagnetic actuators for six pins. The electromagnetic actuators are build by modifying electromagnetic relay generally used as switch in electrical circuits. In the "Fig. 5", it is showing the actuators build from electromagnetic relay.

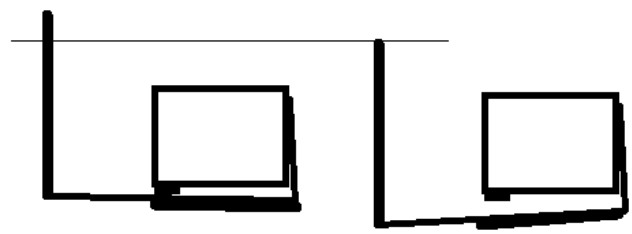

Fig. 5: Fixed length linear actuator. Normal state(right),actuated(left)

Placing such actuators in the a way make the 6 dots placed in standard size. "Fig. 6", shows the single cell consist of 6 actuators.

Fig. 6: Single cell

"Fig. 7", shows the device output while raising 5 dots to show some symbol.

Fig. 7: 5 dots are raised

C. Electronic control system

This system provide necessary connections and power to several unit including a processor to control these units. “Fig. 8”, shows the the system where the red lines shows the power supply connections and the blue line is the data channels. In the heart of the system there is a microcontroller which is actually Arduino-mega board that controls all these units to make the device as a book reader. The microcontroller have 8KB RAM, 2KB EEPROM and runs on 16MHz. There is driver circuit for the cell consists of six actuators described above made of 6 NPN transistors. The microcontrollers send signals to those transistors to make actuation in the linear actuators. A SD card module that support micro SD card was used to enable the microcontroller able to read files. An electronic buzzer was used to make audio sound. There is a button interface consist of 4 buttons and along with the buzzer it provide interactive sound interface controlled by the processor to get input from user in a compact way. To make the device portable a two cell Lipo battery was used power the device. The USB port provide an interface to update firmware when necessary. To store information permanently we have used the EEPROM included in the microcontroller.

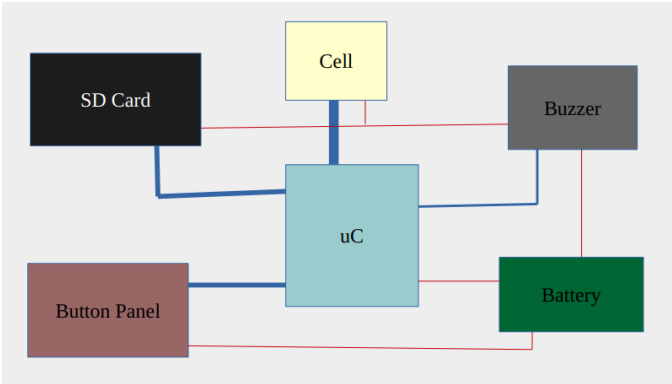

Fig. 8: Electroncis System

D. The device

The hardware and software part was designed concisely to make it easy to use. There is a menu system that can be navigated using the button interface. “Fig. 9”, shows the button interface along with the cell position in the device. The are three menus represented by the sound interfaces shown bellow.

1) Brows Book (long beep, beep , beep)
2) Read from last (beep, long beep, beep)
3) Tutorial (beep, beep, long beep)

By selecting menus user can read new book, read from last saved lines and run the tutorial. There are some defined sound for events like ”enter”, ”success”, ”error” etc. Using this interactive interface user can easily operate the device. New book can inserted by taking the SD card and copying file from computer. Our PC software can convert any book written in Unicode Bangla in text file to the braille book format we introduced before.

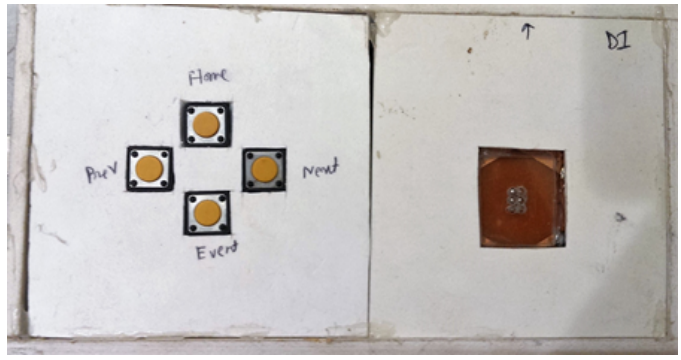

Fig. 9: Interfaces

Experiment and Results

We have built several devices and asked the visually impaired persons to read using this device. Fig[9], Fig[10] and Fig[11] shows some user reading through the device.

Fig. 10: Visually impaired student reading through the device

Fig. 11: Reading experiment

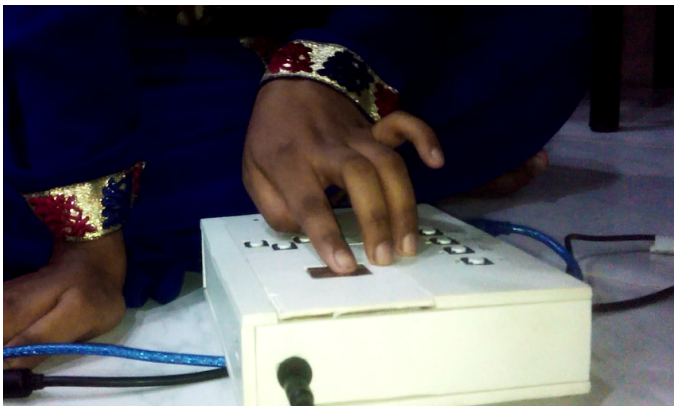

Fig. 12: Reading experiment three

We kept three braille converted Bangla book each of having more than hundred of pages. We took the device to the Ekushe boimela and there were a book stall for the visually impaired people where we demonstrated the devices. In that time around 10 users read books through the device and they were very excited to be able to read book. That day two users read around 10 pages each of two different books.

CONCLUSION

Refreshable braille book reader which is able to read Bangla book can play a vital role in the education sector of Bangladesh. Here from the experiment we found visually people were able to read books through this device and have some pros and cons also as having only single cell. Like they have to remember the previous symbol as some alphabets require two cell to represent. From the users who were mostly student we found very much positive feedback to bring the device in market.

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